DEMO PACK

OpenID Vision

Know when to match. Know when to reject.

How to run the system, what you will see, and how to read every screen - for people who did not build it.

In this pack

- **This guide** - set-up, every scenario with the real output, how to read the screen
- **JURY_DECK.pdf** - 14 slides · **JURY_GUIDE.pdf** - what to say, demo script, 30 questions with honest answers
- **samples/** - public-domain pictures and clips with known right answers (and `SOURCES.md`)
- **run_demo.py** - runs every scenario and prints PASS / FAIL · **face_check.py** - try your own volunteers' photos
- **expected_output/** - the real results of a reference run · **screenshots/** - the real screens

Three commands

```
powershell -ExecutionPolicy Bypass -File scripts\setup.ps1      # once
powershell -ExecutionPolicy Bypass -File scripts\start.ps1      # starts the system on
http://127.0.0.1:8000
python demo\run_demo.py --compare                              # runs every scenario and
shows the results
```

Independent technical prototype responding to the Zed Digital PS-1 assignment - not an official Zed Digital product. Research use with consenting people only. Photos in this pack are public domain; heads in the street photos are blurred.

1. What you are about to see

The system answers one question about a person in a photo: **"is this somebody who was enrolled?"** It gives one of three answers, and always says why.

MATCH

"This is enrolled person X." The evidence passed an explicit rule.

UNKNOWN

"Nobody enrolled is supported by the evidence." Not an accusation; unknown people are never saved or tracked.

REVIEW

"Not sure - a person should look." Poor photo, no usable face, or a borderline score.

Why it works differently from a clothing-based matcher

It looks at the **face first**. When a usable face is visible the face decides; clothing and build are shown as supporting evidence but **can never turn a face that does not match into a MATCH**. When no usable face is visible it uses the body - and, because that is weaker evidence, asks a person (REVIEW) instead of matching automatically.

The numbers to remember. Face-stage accuracy on the standard LFW test: **98.9 %** (6,000 pairs). Automatic-MATCH threshold chosen so that strangers reach it about **1 time in 100,000**. In our end-to-end test a clothing-only matcher accepted **51 of 60** strangers; face + body accepted **0**.

Set-up (Windows 10/11, once)

```
cd D:\ps1
powershell -ExecutionPolicy Bypass -File scripts\setup.ps1          # creates the Python
environment, builds the interface, downloads and verifies the models
powershell -ExecutionPolicy Bypass -File scripts\start.ps1          # leave this window open; then
open http://127.0.0.1:8000
```

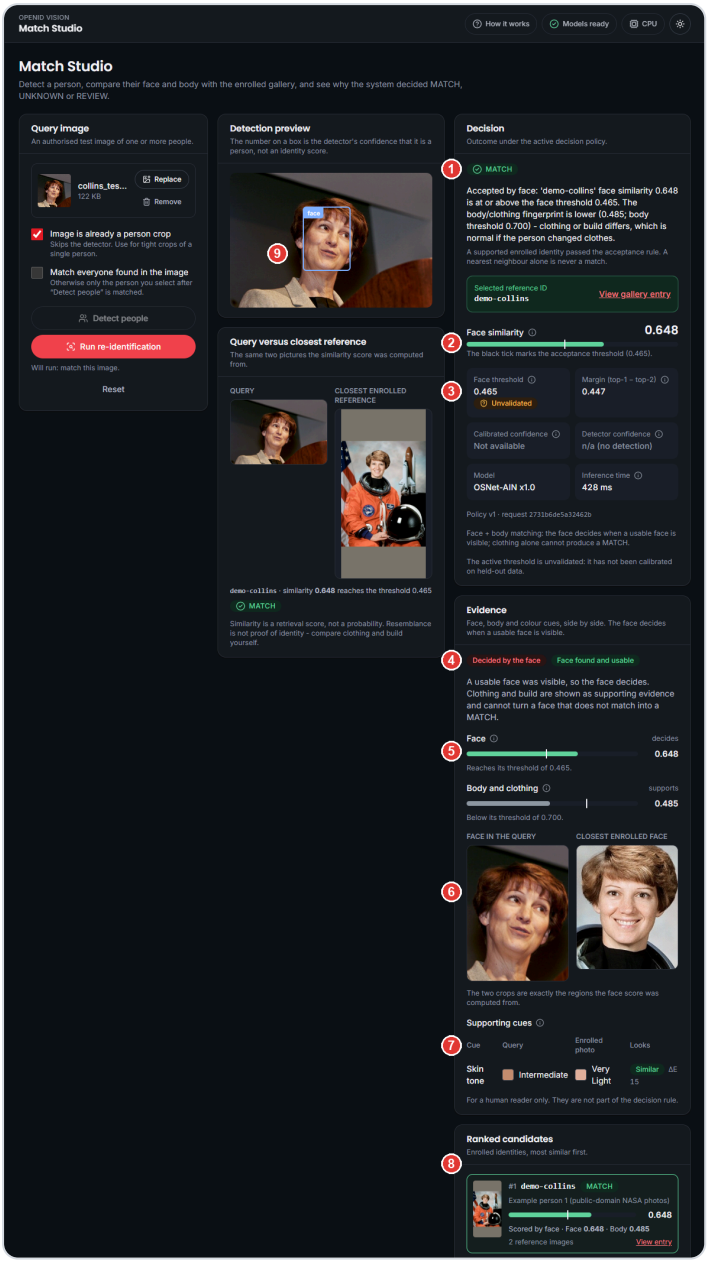
The top bar should say **Models ready · CPU**. If it does not, see *What can go wrong* (section 8). Nothing is downloaded while the system runs; the model files are checked against published checksums.

Run every scenario with one command

```
python demo\run_demo.py --compare         # about 1 minute; add --lab for the Lookalike Failure Lab
(1-2 more minutes)
```

It enrols three example people, shows the system the pictures and clips in section 3, and prints for each: what the answer **should** be, what it **was**, and why. The reference run is in `expected_output/`; yours should match (scores may differ in the last decimals on other hardware). The example people stay enrolled so you can repeat any scenario by hand. `python demo\run_demo.py --cleanup` removes them.

2. How to read the result screen (Match Studio)



1. **The answer** (MATCH / UNKNOWN / REVIEW) and a sentence built from the real numbers.
2. **Face similarity** against its threshold (the black tick). A bar that reaches the tick passes. It is a score, not a percentage.
3. **Threshold** and the policy status. *Unvalidated* = a starting value not yet calibrated on this system's own cameras.
4. **Which evidence decided**: the face, or - when no usable face was visible - the body.
5. **Both scores side by side.** Face = how alike the faces are; Body and clothing = supporting evidence (it can be low when the person changed clothes).
6. **The two faces** the face score was computed from: query on the left, closest enrolled on the right. Judge it yourself.
7. **Colour cues** (hair, skin tone, clothing). For a human reader only; they never change the decision.
8. **All candidates**, each with its face and body score. The nearest one is shown for transparency; it is only a MATCH if it passed the rule.
9. **The face the system found**, drawn on the query picture.

Tip: the **How it works** button in the top bar opens a plain-language tour of the whole system inside the application.

3. The pictures in this pack

Every picture has a known right answer. Faces: official NASA photographs (public domain). Street scenes: CC0 photos with every head blurred. Licences and how each file was made: `samples/SOURCES.md`.

Enrolled (shown to the system first, two photos each)



Example person 1 - flight suit, 1998



Example person 1 - navy jacket



Example person 2 - flight suit, 1992



Example person 2 - blue jacket, 1987



Example person 3 - street photo, head blurred (no face on file)



Example person 3 - mirror image

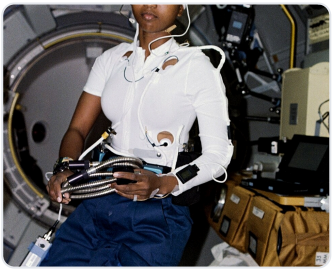
Shown afterwards (new photos)



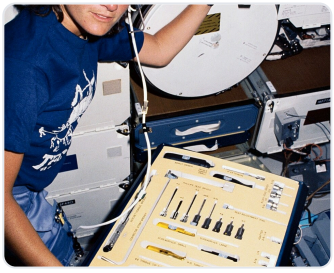
person 1, 1999 → should be MATCH



person 1, 2010 → MATCH



person 2, in flight → MATCH



a stranger → UNKNOWN



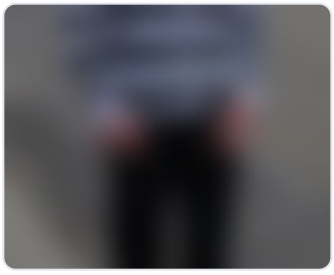
same stranger → UNKNOWN



person 3, no face visible → REVIEW



a stranger, no face → UNKNOWN



person 3, out of focus → REVIEW

Clips: `clip_faces.avi` (person 1 · stranger · person 1) and `clip_street.avi` (person 3 · other scene · person 3). Both are 3 seconds long; Video Explorer takes one frame per second and judges each frame on its own - it does **not** track people.

4. The scenarios, with the real output

Each card shows the picture, how to repeat it by hand, and what the system actually answered in the reference run (read from `expected_output/`).



Scenario 1: Same person, different clothes and 12 years later

Do: Match Studio → tick *Image is already a person crop* → upload → *Run re-identification*

Result: **MATCH** **collins** decided by **face** · face **0.648** · body **0.605** · threshold 0.465

> Accepted by face: 'demo-collins' face similarity 0.648 is at or above the face threshold 0.465. The body/clothing fingerprint is lower (0.605; body threshold 0.700) - clothing or build differs, which is normal if the person changed clothes.



Scenario 2: Same person, a different setting

Do: same steps

Result: **MATCH** **collins** decided by **face** · face **0.624** · body **0.570** · threshold 0.465

> Accepted by face: 'demo-collins' face similarity 0.623 is at or above the face threshold 0.465. The body/clothing fingerprint is lower (0.570; body threshold 0.700) - clothing or build differs, which is normal if the person changed clothes.

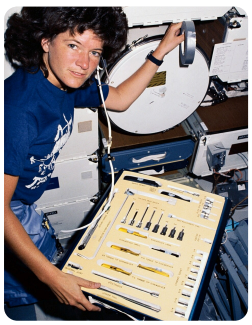


Scenario 3: A second enrolled person

Do: same steps - the system must pick the RIGHT enrolled person

Result: **MATCH** **jemison** decided by **face** · face **0.572** · body **0.434** · threshold 0.465

> Accepted by face: 'demo-jemison' face similarity 0.572 is at or above the face threshold 0.465. The body/clothing fingerprint is lower (0.434; body threshold 0.700) - clothing or build differs, which is normal if the person changed clothes.



Scenario 4: A stranger

Do: same steps - nobody enrolled looks like this person

Result: **UNKNOWN** decided by **face** · face **0.261** · body **0.580** · threshold 0.465

> No supported match: a usable face was compared with 2 enrolled face holder(s); the closest, 'demo-collins', scored 0.261, below the face threshold 0.465.



Scenario 5: Another stranger

Do: same steps

Result: **UNKNOWN** decided by **face** · face **0.211** · body **0.593** · threshold 0.465

No supported match: a usable face was compared with 2 enrolled face holder(s); the closest, 'demo-collins', scored 0.211, below the face threshold 0.465.



Scenario 7: Enrolled street person, but no face visible

Do: same steps - the head is blurred, so only clothing can be compared

Result: **REVIEW** decided by **body** · face **-** · body **0.936** · threshold 0.700

No face was found in the picture. Body/clothing evidence alone: 'demo-person-a' scored 0.936, at or above the body threshold 0.700, but clothing and build are not enough for an automatic MATCH - a person should confirm.

Face: No face was found in this picture, so only the body could be compared.



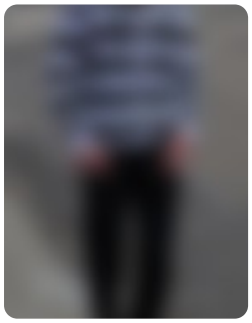
Scenario 8: A stranger without a visible face

Do: same steps

Result: **UNKNOWN** decided by **body** · face **-** · body **0.500** · threshold 0.700

No supported match: the closest candidate 'demo-jemison' scored 0.500, below the acceptance threshold 0.700. No face was found in the picture.

Face: No face was found in this picture, so only the body could be compared.



Scenario 9: A poor-quality picture

Do: same steps

Result: **REVIEW** decided by **body** · face **-** · body **0.604** · threshold 0.700

Image quality is inadequate for an automatic decision (blurry). Closest candidate 'demo-person-a' scored 0.604. A face was found but it is not good enough to compare (face_low_confidence).

Face: A face was found but it is not usable: The face detector was not sure this is a face.

4. The scenarios (continued): video and a busy scene

Scenario 6: video `clip_faces.avi`

Do: Video Explorer → Browse files → 1 s → Analyse video

Result per sampled frame: **t=0s MATCH → t=1s UNKNOWN → t=2s MATCH**

Frame by frame - person 1, then the stranger, then person 1 again. No tracking.

Scenario 10: a busy scene

Do: Match Studio → tick *Match everyone found in the image* → upload `scene_B_station_square.jpg`

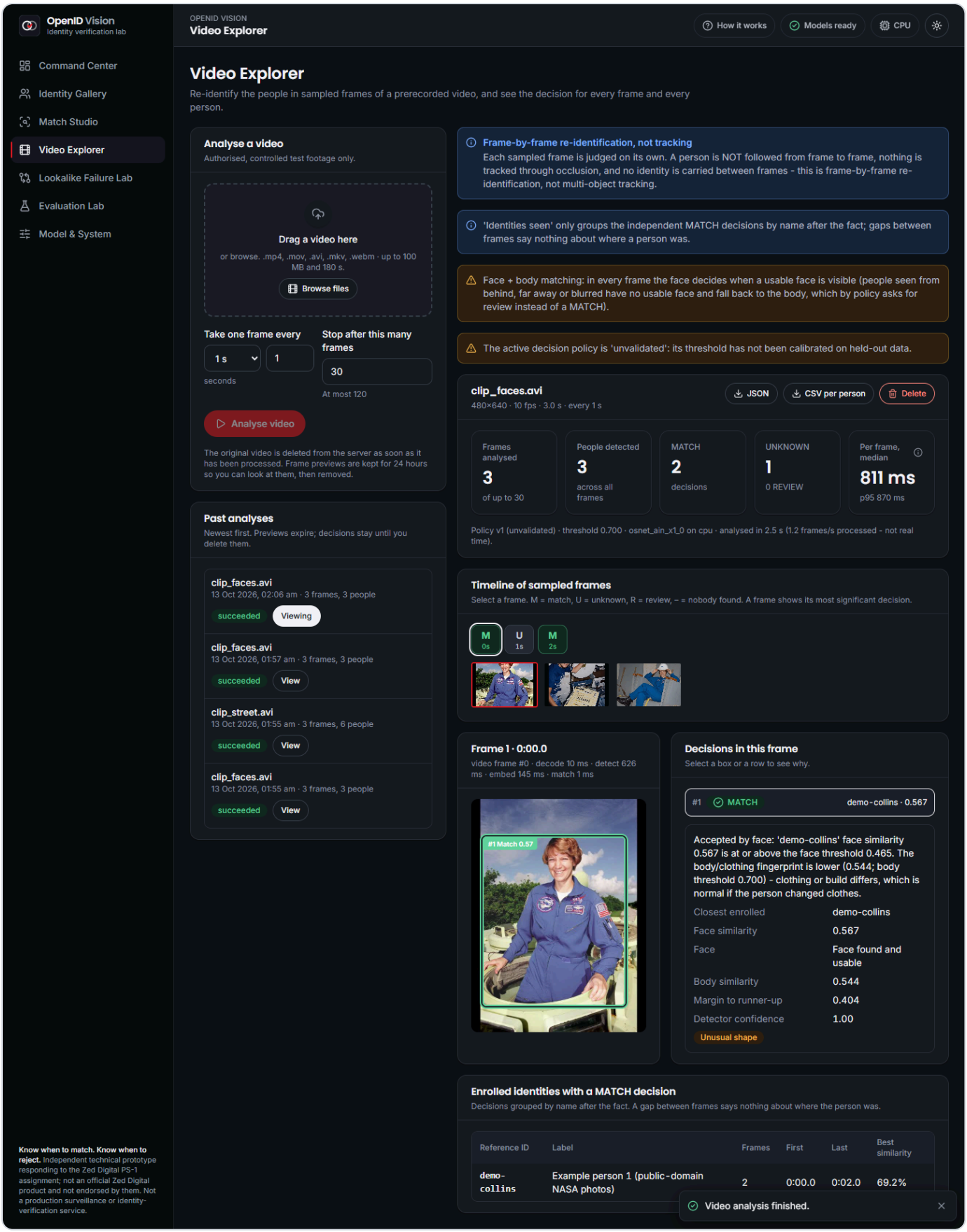
Result: seven people found and decided separately (heads blurred, nobody enrolled): **6 × UNKNOWN, 1 × REVIEW**.

Scenario 11: video `clip_street.avi` (no usable faces)

Do: same as scenario 6

Result per sampled frame: **t=0s REVIEW → t=1s UNKNOWN/UNKNOWN/UNKNOWN/UNKNOWN → t=2s REVIEW**

Clothing matches in frames 1 and 3, but with no face the system asks a person (REVIEW) instead of matching.



Video Explorer shows a timeline (M = match, U = unknown, R = review), the frame with the people drawn on it, the decision and its reason, and measured timings. A video upload is deleted from the server as soon as it has been processed.

5. Clothing only versus face + body

The **same eight photos** were shown to the system twice: once under the original clothing-only policy, once under face + body. *(Created with `run_demo.py --compare`, which switches the policy and puts it back.)*

Photo	Should be	Clothing only	Face + body
collins_test_1_nasa_jacket_1999	MATCH collins	UNKNOWN (wrong)	MATCH collins
collins_test_2_suit_jacket_2010	MATCH collins	UNKNOWN (wrong)	MATCH collins
collins_test_3_hard_hat_2004	MATCH collins	UNKNOWN (wrong)	MATCH collins
jemison_test_1_in_flight_1992	MATCH jemison	UNKNOWN (wrong)	MATCH jemison
jemison_test_2_lab_1992	MATCH jemison	UNKNOWN (wrong)	MATCH jemison
ride_stranger_1_cockpit_1983	UNKNOWN	UNKNOWN	UNKNOWN
ride_stranger_2_blue_shirt_1983	UNKNOWN	UNKNOWN	UNKNOWN
ride_stranger_3_headset_1983	UNKNOWN	UNKNOWN	UNKNOWN

Right answers: clothing only 3 of 8 · face + body 8 of 8. The same person in different clothes cannot be recognised by clothing; the face recognises her across clothes and years.

The opposite failure - same clothes, different person - measured on 10 enrolled people and 60 strangers

	Clothing only	Face + body
Strangers wrongly **accepted** (lower is better)	**51 of 60**	**0 of 60**
Enrolled people recognised correctly (higher is better)	**5 of 20**	**20 of 20**
Enrolled people matched to the **wrong** person	**8**	**0**

Read these honestly. The 60 strangers come from LFW (public-figure news photos, head and shoulders) - outside what the clothing model was designed for, so its column is a worst case. The threshold was fixed beforehand, not tuned on these photos. A small sample: it shows *what the face adds*; it is not a benchmark. Reproduce: `python scripts\verify_face_vs_body.py --lfw <folder>`.

6. Try it with your own volunteers

The strongest demonstration uses people in the room (with their permission). No web interface needed:

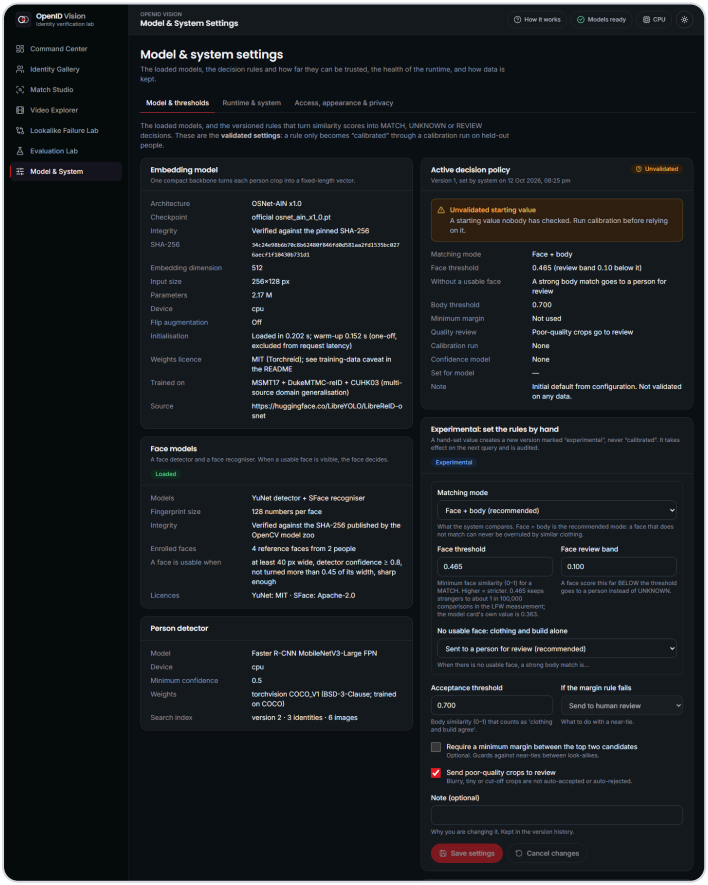
```
my_photos\
   enroll\anna\   anna_1.jpg  anna_2.jpg        (2-3 clear, front-facing photos of each person to
enrol)
   enroll\ben\    ben_1.jpg   ben_2.jpg
   test\anna\     new_photo.jpg  back_view.jpg   (NEW photos of enrolled people - should be MATCH,
or REVIEW if poor)
   test\visitor\  visitor_in_similar_clothes.jpg  (somebody NOT enrolled - should be UNKNOWN)

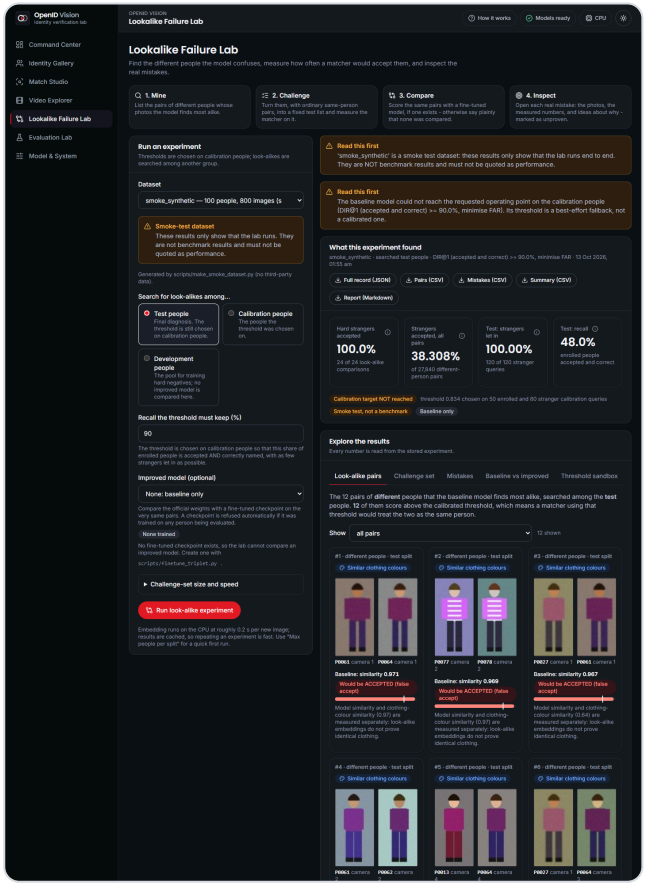
python demo\face_check.py my_photos --i-have-consent
```

It prints, for every test photo, what the answer should be, what the system said, and the face and body scores; it removes the volunteers again at the end (`--keep` leaves them). Try the public-domain set first: `python demo\face_check.py demo\samples\faces --i-have-consent`.

> **Best live demonstration:** enrol a volunteer, then show a photo of a *different* person wearing similar clothes. The old clothing-only matcher accepts them; this one says UNKNOWN, and the Evidence card shows why (a high body bar, a low face bar).

7. A short tour of the other screens





Model & System - models with checksums, the active rules (versioned, nothing edited in place), the switch between Face + body and Body only.

Lookalike Failure Lab - finds the different people the model confuses; every cause is labelled a hypothesis. (Synthetic data in the screenshot: proves the lab works, not accuracy.)

8. What can go wrong

You see	Why and what to do
Top bar: **Backend offline**	The server is not running. Start it with `scripts\start.ps1` and keep that window open.
Top bar: **Model not loaded**	Model files missing or damaged. Run `python scripts\download_weights.py`, restart. The reason is shown under Model & System.
"Face models are not loaded - only the body is compared"	The two face files are missing. Same fix. Until then the system works on the body only and says so.
`run_demo.py` says **BODY ONLY**	The policy is not Face + body. Run with `--enable-faces`, or choose it under Model & System → Matching mode.
A scenario result differs from this guide	Other people in the gallery, a changed threshold, or a different policy can change results. Use a clean gallery (`run_demo.py --cleanup`) and policy v1.
A person card says **No face on file**	The reference photos have no usable face (back view, tiny, blurred). Add a clear front-facing photo; or press **Analyse faces**.
Port 8000 is in use	Use `scripts\start.ps1 -Port 8020` and pass `--base-url http://127.0.0.1:8020` to the scripts.
Scores differ slightly from the guide	Normal across different CPUs. Outcomes should be the same.

9. What this pack does not prove

- It is **not a phone's Face ID**: no depth sensor and no liveness check; a photograph can fool a 2-D matcher.
- The face stage was measured on the **standard LFW benchmark** (easy). Other cameras, lighting, ages and demographic groups were **not** measured.
- The **Market-1501** benchmark ran on a CPU-sized subset with the project's own protocol, **body only**: Rank-1 83.6 %, false-accept rate 6.4 / 12.9 / 28.6 % at 50 / 60 / 70 % recall, a 90 % target unreachable. Face + body was **not** benchmarked on pedestrian data.
- Thresholds are evidence-based defaults, **unvalidated** for any specific camera.
- The example people are public figures photographed at work; they are used only to show recognition across photographs.
- Face fingerprints are **biometric data**: enrol only consenting people; deleting a person erases everything stored about them.

10. Where everything is

`docs/JURY_GUIDE.md`	what to say, demo script, questions and answers

`docs/FACE_AND_BODY_MATCHING.md`	the exact rule, the threshold derivation, all measurements and limits
`docs/SECURITY_AND_PRIVACY.md`	consent, biometric data, retention, security
`docs/results/*.json`	the raw measurements behind every number in this pack
`README.md`	installation, API, evaluation, troubleshooting